NEXASTORE

AI Machines • Developer Workstations • Next-Generation Computing



LOCAL LLM

Local LLM Hardware Requirements Guide

Plan RAM, VRAM, storage and compute for local models

NexaStore Knowledge Series • 2026 Edition

1. What Determines Local LLM Requirements?

Model parameter count is only one factor. Precision or quantization, context length, runtime, KV cache, concurrency and CPU/GPU offloading all affect memory requirements. The figures in this guide are planning ranges for educational comparison, not guarantees for a particular model.

Model class	Practical memory direction	Suggested hardware profile
1B-3B	Low memory footprint; often suitable for modest systems	16-32 GB RAM; GPU optional depending on runtime
7B-8B	Common local-development tier when quantized	32 GB RAM; 8-12 GB VRAM useful
13B-14B	Requires noticeably more memory/headroom	64 GB RAM; 12-16+ GB VRAM preferred
30B-34B	High local memory demand	64-128 GB RAM; 24 GB+ VRAM or CPU/GPU split
70B-class	Very high local requirements	128 GB+ RAM and high-VRAM workstation; cloud may be more practical

2. Quantization Changes the Equation

Concept	Impact
Full / high precision	Higher memory demand; useful in some research/training scenarios.
8-bit quantization	Reduces memory relative to higher precision while preserving strong quality for many use cases.
4-bit quantization	Common for local inference because it can substantially reduce model memory requirements.
Context length	Longer contexts increase runtime memory beyond the static model weights.
Concurrency	Multiple simultaneous requests increase memory and compute requirements.

3. RAM, VRAM and Storage

Resource	Role in local LLM workloads
GPU VRAM	Fast memory used by GPU-resident model weights, activations and runtime state. Often the primary constraint for GPU inference.
System RAM	Supports CPU inference/offload, application services, model loading, containers and databases.
NVMe storage	Stores model files, caches and datasets. Models can consume many gigabytes each.
CPU	Important when models run partly or entirely on CPU and for surrounding application work.
NPU	Useful for supported on-device workloads but local LLM support varies by model/runtime and should be verified.

4. Example Development Profiles

Profile	RAM	VRAM	Example intent
Starter	32 GB	8 GB	Small quantized models and learning
Serious local developer	64 GB	12-16 GB	7B/14B-class experimentation plus development stack
Advanced workstation	128 GB	24-32 GB	Larger local models and heavier concurrent AI services
High-end lab	128-256 GB	32 GB+	Large local experiments; cloud still useful for scale

5. Why Exact Answers Need Runtime Context

A question such as 'Can this laptop run a 14B model?' cannot be answered from parameter count alone. The agent should consider model format, quantization, context size, desired tokens per second, GPU support, available VRAM, system RAM and whether CPU offloading is acceptable.

User requirement	What the recommendation agent should inspect
Run a 14B model locally	Exact SKU RAM/VRAM + this guide + model/runtime requirements
Need fast responses	GPU capability, VRAM and expected runtime acceleration
Need long context	Available memory headroom and runtime context support
Run model plus Docker stack	System RAM, CPU, GPU VRAM and background service footprint
Occasional local use only	Balance hardware cost against cloud inference alternatives

Educational notice

Memory ranges are intentionally approximate for teaching architecture and recommendation reasoning. They are not compatibility guarantees for third-party models.